

A comparison between OpenNebula, OpenStack and CloudStack

Feature/s	OpenNebula	OpenStack	CloudStack
Cloud APIs and user interface	OCCL and EC2 API	NOVA API compatible with EC2 (Elastic Compute Cloud) and S3 AWS	Amazon Web Services
Development model	Public development	Public development	Public development
Production readiness	Enterprise-ready and direct support from developers	No, only available through any of the several vendor-specific stacks	Enterprise-ready and also has direct support from developers
Programming language	Java and Ruby	Python	Java
Networking model	VLAN	Flat Flat DHCP VLAN DHCP	VLANs, security groups

- **APIs and interfaces:** These are used to manage and monitor OpenNebula components. [To manage physical and virtual resources, they work as an interface.
- **Users and groups:** These support authentication, and authorise individual users and groups with the individual permissions.
- **Hosts and VM resources:** These are a key aspect of a heterogeneous cloud that is managed and monitored, e.g., Xen, VMware.
- **Storage components:** These are the basis for centralised or decentralised template repositories.
- **Network components:** These can be managed flexibly. Naturally, there is support for VLANs and Open vSwitch.

The front-end

- The machine that has OpenNebula installed on it is known as the front-end machine, which is also responsible for executing OpenNebula services.
- The front-end needs to have access to the image repository and network connectivity to each node.
- It requires Ruby 1.8.7 or above.
- OpenNebula's services are listed below:
 - Management daemon (Oned) and scheduler (mm_sched)
 - Monitoring and accounting daemon (Onectcd)
 - Web interface server (Sunstone)
 - Cloud API servers (EC2- query or OCCL)

Virtualisation hosts

- To run the VMs, we require some physical machines, which are called hosts.
- The virtualisation sub-system is responsible for communicating with the hypervisor and taking the required action for any node in the VM life cycle.
- During the installation, the admin account should be enabled to execute commands with root privileges.

Storage

Data stores are used to handle the VM images, and each data store must be accessible by the front-end, using any type of storage technology.

- OpenNebula has three types of data stores:
 - File data store – used to store the plain files (not disk images)

- Image data store – repository for images only
- System data store – used to hold the running VM images

The image data store type depends on the storage technology used. There are three types of image data stores available:

- File system – stores VM images in file formats
- LVM – reduces the overhead of having the file system in place; the LVM is used to store virtual images instead of plain files
- Ceph – stores images using Ceph blocks

OpenNebula can handle multiple storage scenarios, either centralised or decentralised.

Networking

There must be at least two physical networks configured in OpenNebula:

- Service network - to access the hosts to monitor and manage hypervisors, and to move VM images.
- Instance network - to offer network connectivity between the VMs across the different hosts.

Whenever any VM gets launched, OpenNebula will connect its network interfaces to the bridge described in the virtual network definition.

OpenNebula supports four types of networking modes:

- Bridged—where the VM is directly attached to the physical bridge in the hypervisor.
- VLAN—where the VMs are connected by using 802.1Q VLAN tagging.
- Open vSwitch—which is the same as VLAN but uses an open vSwitch instead of a Linux bridge.
- VXLAN—which implements VLAN using the VXLAN protocol. 

References

- [1] https://www.researchgate.net/publication/268747387_OpenNebula_Open_Source_IaaS_Cloud_Computing_Software_Platforms
- [2] http://docs.opennebula.org/5.0/intro_release_notes/

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